

Project Proposal

This project investigates the automated generation of natural language ecological risk assessment reports from raw Google Earth Engine (GEE) tabular data. Specifically, I will focus on translating a core set of remote sensing and meteorological metrics—NDVI, elevation, slope, temperature, and wind speed—into accessible, human-readable analytical text. This problem is highly interesting because while Large Language Models (LLMs) excel at natural language generation, they frequently hallucinate or fail to accurately capture the physical correlations between multiple numerical variables, such as understanding that high wind speed combined with a steep slope and low NDVI collectively indicates severe wildfire vulnerability. To provide context and theoretical background, I will examine literature on data-to-text generation, tabular data reasoning within Transformer architectures, and prompt engineering techniques for mitigating numerical hallucinations. The dataset will consist of regional environmental data extracted via the GEE API, which I will dynamically collect and format into structured JSON or Markdown tables for the model to process. My proposed method involves utilizing a pre-trained open-source LLM, such as Llama 3 or Mistral, accessed via the Hugging Face transformers library. While I will rely on these existing foundational models, I plan to modify the generation pipeline by designing specialized few-shot prompts and potentially exploring lightweight fine-tuning techniques (like LoRA) to strictly guide the model's ecological reasoning. My general approach is to first structure the raw GEE data, construct dynamic prompts embedding this tabular information, and then instruct the model to synthesize a coherent analytical report. To evaluate the results qualitatively, I expect to produce comparative text samples that demonstrate the model's ability to accurately describe environmental vulnerabilities without fabricating statistics. Quantitatively, I will evaluate the generated reports against human-written baseline summaries using standard NLP performance metrics such as ROUGE and BLEU, alongside a strict exact-match factual consistency algorithm to ensure the numbers in the text perfectly align with the input data. Because this work is combined with my senior graduation project, the unique portion counting toward this LLM class will strictly be the NLP pipeline: tabular data formatting for LLM ingestion, prompt engineering, model implementation, and the comprehensive text evaluation, whereas the raw GEE data extraction belongs exclusively to the graduation project.